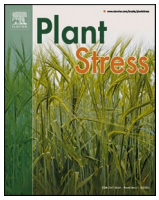
Plant Stress 14 (2024) 100565



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Plant Stress

journal homepage: www.sciencedirect.com/journal/plant-stress





Cultivating resilience: Harnessing pyoverdine-producing *Pseudomonas* to contrast iron deficiency in cucumber plants

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ARTICLE INFO

Keywords:
Pseudomonas
Plant growth-promoting bacteria
Pyoverdine
Fe deficiency
FCR activity
FRO genes

ABSTRACT

Iron (Fe) deficiency in crops significantly reduces yield, impacting agricultural productivity worldwide. Synthetic Fe chelates are commonly applied as fertilizers to address this issue, but their synthetic nature and prolonged use poses environmental risks. Thus, inoculation of plant growth-promoting bacteria rises as an alternative to enhance Fe uptake in crops while minimizing reliance on synthetic chelates. This study aimed to examine the influence of *Pseudomonas* RMC4 inoculation and pyoverdine application on cucumber plants cultivated hydroponically under Fe deficiency conditions. Evaluations included the SPAD index, plant biomass, root morphology, Fe-chelate reductase activity, gene expression, and ionomic analysis. Following Fe deficiency, *Pseudomonas* RMC4 inoculation improved the SPAD index, increased dry weight, enhanced root development, and facilitated Fe acquisition mechanisms, thus, fostering the endogenous resilience of the plant to the limited Fe availability. This improvement was observed with bacterial inoculation or pyoverdine application alongside an insoluble Fe source (ferrihydrite). Overall, the results suggest the beneficial impact of *Pseudomonas* RMC4 inoculation in alleviating symptoms of Fe deficiency. Future studies will investigate bacterial application under field conditions to assess its potential in replacing synthetic Fe-chelates fertilizers in crop production in favor of more sustainable agriculture.

1. Introduction

Iron (Fe) is an essential micronutrient for plants, being thus required to guarantee equilibrated growth and optimal productivity (Kobayashi and Nishizawa, 2012). Despite being widely distributed on the Earth's crust, its availability for plant uptake can be limited due to different chemical and physical characteristics of soil, thus often inducing the onset of Fe chlorosis phenomena (Mimmo et al., 2014). Problems associated with Fe deficiency especially affect widely cultivated horticultural crops, such as cucumber (Bacaicoa and García-Mina, 2009), being therefore the most critical issue for these crops grown on soils with low availability of Fe (such as alkaline and calcareous soils) (Álvarez-Fernández et al., 2011; Lucena et al., 2007). The application of synthetic Fe chelates has represented the most common solution to overcome the nutritional disorder in horticultural crops. These molecules, based on the Fe chelation with aminocarboxylate ligands (e.g., ethylenediaminetetraacetic acid (EDTA), diethylenetriaminepentaacetic acid (DTPA), and ethylenediamine-N,N′-bis(hydroxyphenylacetic) acid (EDDHA)), are very effective in maintaining the metal in the soil solution, and, consequently, increasing its availability for root acquisition (López-Rayo et al., 2019). However, due to their low biodegradability, high cost, and increased soil accumulation, the need for more environmentally friendly Fe fertilizers has nowadays become particularly crucial (Zuluaga et al., 2023a). In this context, the application of biostimulants might represent an interesting alternative for developing Fe fertilizers in a vision of more sustainable agriculture (Corsi et al., 2022). Biostimulants are substances, mixtures, or microorganisms that positively affect plant growth as well as stimulate nutrients acquisition

Abbreviations: ACCD, 1-aminocyclopropane-1-carboxylate deaminase; DAP, day after planting; DAT, day after treatment; DTPA, diethylenetriaminepentaacetic acid; EDTA, ethylenediaminetetraacetic acid; FCR, ferric-chelate reductase; IAA, indole-3-acetic acid; ICP-OES, Inductively coupled plasma optical emission spectroscopy; LB, Luria-Bertani; MMS, Minimal medium succinate; NS, nutrient solution; o-o-EDDHA, ethylenediamine-N-N′-bis(*o*-hydroxyphenylacetic) acid; PGPB, plant growth-promoting bacteria; RDW, root dry weight; SDW, shoot dry weight.

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https://doi.org/10.1016/j.stress.2024.100565
Received 26 June 2024; Received in revised form 2 August 2024; Accepted 14 August 2024
Available online 15 August 2024
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processes (du Jardin, 2015). With respect to Fe nutrition, biostimulants can potentially enhance Fe nutritional state of plants, although, they do not directly supply this micronutrient. Some examples in this sense include beneficial microorganisms, also known as Plant Growth-Promoting Bacteria (PGPB), silicon, algae extracts, or protein hydrolysates among others (Celletti et al., 2020; González-Pérez et al., 2021; Gonzalo et al., 2013; Zuluaga et al., 2021).

The use of PGPB in agriculture has been extensively studied due to their wide range of applications, including i) the control of pathogens through the secretion of compounds like HCN (Haas and Défago, 2005), ii) the stimulation of plant growth by the secretion of phytohormones such as indole-3-acetic acid (IAA), 1-aminocyclopropane-1-carboxylate deaminase (ACCD), or cytokinins (Ferreira et al., 2019), and iii) the enhancement of root nutrient uptake through the secretion of siderophores by the bacteria (Soares, 2022). Some of the bacteria used as PGPB include members of the genera *Enterobacter* (Zuluaga et al., 2021), *Azotobacter, Bacillus* (Ferreira et al., 2019) or *Pseudomonas* (Vindeirinho et al., 2021).

In particular, the genus of *Pseudomonas* encompasses Gram-negative bacteria, comprising several species identified as PGPB. The dominant PGP traits exhibited by these bacteria include phosphate solubilization (Miller et al., 2010), nitrogen fixation (Mirza et al., 2006), ammonia production (X. Yang et al., 2012), production of ACC deaminase (Saravanakumar and Samiyappan, 2007), production of phytohormones (Reetha et al., 2014) and siderophores (Meyer and Abdallah, 1978). The beneficial properties of bacteria belonging to the genus *Pseudomonas* for plant growth have already been investigated. For instance, Esitken et al. (2010) performed an experiment using *Bacillus* M3, *Bacillus* OSU-142, and *Pseudomonas* BA-8 in strawberry field production for three years observing that root inoculation with *Pseudomonas* significantly increased yield (10,5 %), and the root inoculation of *Bacillus* M3 in combination with *Pseudomonas* BA-8 increased the yield, phosphorous and zinc content, and development under field conditions. Sandini et al. (2019) compared the inoculation of maize seeds with *P. fluorescens* or *Azospirillum brasilense*. The authors found that plants inoculated with *P. fluorescens* increased yields by 29.5 %, even when the recommended dose of N fertilizer was reduced by 25 % while no significant differences were observed in plants inoculated with *A. brasilense*.

Several strains of *Pseudomonas* are also known for their production of siderophores, described as low molecular weight compounds with a high affinity for Fe^{3+}(Hider and Kong, 2010). Several species within the *Pseudomonas* genus, including *P. aeruginosa, P. chlorophis, P. fluorescens*, and *P. putida* have been described to produce a siderophore called pyoverdine. The Fe complexation with pyoverdine presents very high stability (stability constant $\log K^{\circ}_{FeL} = 31.06$ (Boukhalfa et al., 2006)), even higher than the EDTA-Fe ($\log K^{\circ}_{FeL} = 27.6$ (López-Rayo et al., 2019)), but lower than most used commercial Fe fertilizer in calcareous soils like the *o,o*-EDDHA-Fe ($\log K^{\circ}_{FeL} = 37.7$ (Yunta et al., 2003)). The high affinity of pyoverdine for Fe, and its high stability constant when complexed with Fe, could benefit crops when Fe is not available in the soil (in the form of Fe oxides). In this context, pyoverdine has proved itself able to solubilize Fe oxides (Robin et al., 2008). However, evidence showing that this Fe source can be utilized by roots under Fe deficiency conditions is still lacking. Interestingly, Vansuyt et al. (2007) demonstrated that *Arabidopsis thaliana* plants treated with Fe-pyoverdine could acquire Fe from the complex. Nevertheless, the uptake of the micronutrients was not dependent on the activation of the root mechanisms underlying the Fe acquisition system (i.e., increase in reductase activity or in the expression of *IRT1* transporter), suggesting that the incorporation of Fe-pyoverdine was not related to the Fe-reduction based strategy typically active in dicots (Strategy I plants).

Thus, the aim of the present work was to evaluate and characterize the possible contribution of pyoverdine-producing *Pseudomonas* on Fe-deficient cucumber plants (Strategy I). For this purpose, two trials were carried out. Trial 1 aimed to understand the interaction between cucumber plants and *Pseudomonas* under different Fe nutritional conditions. In Trial 2 it was investigated whether the application of either pyoverdine or *Pseudomonas* was able to solubilize insoluble Fe (from Fe oxides) and whether this Fe source could be then taken up by the roots. Additionally, effects of these applications on the activation of the root mechanisms underlying the Fe acquisition process in Strategy I plants have been also investigated.

2. Materials and methods

2.1. Bacterial strain

Pseudomonas RMC4 (accession number PRJNA1028413) was previously isolated from commercial nursery garden (Lozano-González et al., 2023) and selected by its ability to produce pyoverdine and IAA. The bacterial strain was grown in LB liquid medium while subjected to orbital shaking at 180 rpm and maintained at a temperature of 28 °C for 24 h. Afterwards, the cells were harvested, washed, and re-suspended in a sterile saline solution (0.85% w/v NaCl). This bacterial suspension was used to inoculate cucumber roots via the hydroponic nutrient solution to a final concentration of 1×10^6 cells mL^{-1}.

2.2. Pyoverdine (Pvd) production

Pyoverdine siderophore produced by *Pseudomonas* RMC4 was obtained as described by Lozano-González et al. (2023). Briefly, an Erlenmeyer flask containing 200 mL of Fe-free minimal medium succinate (MMS) was inoculated with 200 μL of 24 h broth pre-culture (1×10^8 cell mL^{-1}). The flask was incubated under orbital shaking at 150 rpm and maintained at a temperature of 30 °C for 48 h. Afterwards, the broth culture was centrifuged (3000× g, 15 min), filtered through a 0.45 μm membrane filter, and quantified according to Lozano-González et al. (2023) by absorbance at 400 nm. The cell-free supernatant containing pyoverdine was used to supplement the hydroponic nutrient solution to a final concentration of 2 μM of pyoverdine and bacteria extract.

2.3. Ferrihydrite (Fh) production

Ferrihydrite was synthesized by precipitating $Fe(NO_3)_3 \cdot 9H_2O$ with KOH to a final pH of 7–8, as described by Schwertmann & Cornell (2008). The ferrihydrite obtained was used to supplement the hydroponic nutrient solution provided in a dialysis tube membrane containing 10 mg of the Fe oxide. The dialysis membrane used was a MEMBRA-CEL MD34 14 × 100, MWCO: 14,000, Pore size: 50 Å (Viskase Companies Inc., Willowbrook, IL, USA).

2.4. Plant material and experimental setup

Cucumber plants (*Cucumis sativus* L. cv. Chinese Long) were cultivated hydroponically in a climate-controlled growth chamber with a 14/10-h light/dark period, 24/19 °C, 250 μmol m^{-2} s^{-1} light intensity, and 70 % relative humidity. Seeds were germinated for 5 days in the dark at 22 °C on filter paper moistened with 0.5 mM $CaSO_4$. Five-day-old seedlings were then planted in pots (10 seedlings per pot) filled with 1.5 L of a full-strength nutrient solution (NS) containing 2 mM Ca$(NO_3)_2$, 0.7 mM K_2SO_4, 0.1 mM KH_2PO_4, 0.1 mM KCl, 0.5 mM $MgSO_4$, 10 μM H_3BO_3, 0.5 μM $MnSO_4$, 0.2 μM $CuSO_4$, 0.1 μM $ZnSO_4$, 0.01 μM $(NH_4)_6Mo_7O_{24}$. Two independent experimental trials were performed as follows:

2.4.1. Trial 1: effect of iron nutrition and Pseudomonas inoculation on cucumber growth and Fe uptake mechanisms

Cucumber plants were cultivated under two nutritional levels of Fe ([Fe-]: deficient and [Fe+]: sufficient supplied in the NS as 80 μM Fe-EDTA) and three inoculant conditions with *Pseudomonas* RMC4 (2I: early inoculation 2 days after planting, 6I: late inoculation six days after

planting, control: non-inoculated plants) (Fig. 1). Each treatment consisted of three biological replicates, with 50 seedlings per replicate (five pots). The age of plants is indicated as days after the planting (DAP), which denotes the days after the beginning of hydroponic cultivation. The experiment lasted for 10 DAP. Plants were collected at the initial time (0 DAP) and later on every day until the end of the experiment (10 DAP). These plants were used to determine Fe (III) reductase activity (FCR, three plants per replicate) and for gene expression analysis (1 plant per replicate), as described below. Plant biomass was recorded only at the end of the experiment (10 DAP), using 2 plants per replicate.

2.4.2. Trial 2: effect of Pseudomonas and its siderophore on iron-starved cucumber plants

To gain further insight into the action of *Pseudomonas* RMC4 and its siderophore (pyoverdine), as well as their performance in the presence of sparingly soluble Fe sources (ferrihydrite), the following experiment was carried out (Fig. 2). A group of cucumber plants (36 pots with 10 plantlets per pot) were grown hydroponically during 14 days in Fe deficient (Fe-) conditions, while another group of plants (6 pots with 10 plantlets per pot) were grown for the same time in Fe sufficient (Fe+, supplied in the NS as 80 μM Fe-EDTA) conditions (Fig. 2). At 14 days after planting (14 DAP), plants were split into groups and treatment imposition was carried out as described in Fig. 2. In detail, a set of 30 pots of Fe-deficient plants continued with the Fe starvation and were treated as follows: a group of 6 pots in which Fe deficiency was kept (treatment: [Fe-] Control); a group of 6 pots in which *Pseudomonas* RMC4 was inoculated in the nutrient solution (NS) at a final concentration of 1×10^6 cell ml^{-1} (treatment: [Fe-] RMC4); a group of 6 pots in which the dialysis tube containing 10 mg of ferrihydrite was added to the NS (treatment: [Fe-] Fh); a group of 6 pots in which *Pseudomonas* RMC4 (1×10^6 cell ml^{-1}) plus ferrihydrite (10 mg) were added to the NS (treatment: [Fe-] RMC4+Fh); and a group of 6 pots in which pyoverdine was added to the NS at a final concentration of 2 μM together with the dialysis tube containing 10 mg of ferrihydrite (treatment: [Fe-] Pvd+Fh). The remaining group of 6 pots of Fe-deficient plants, was supplied (14 DAP) with Fe by adding Fe-EDTA in the NS at a final concentration of 80 μM (treatment: [Fe-] Resupply). The group of 6 pots of plants grown under sufficient Fe conditions during 14 DAP, continue growing under such conditions until the harvest (treatment: [Fe+] Control). Each treatment consisted of three biological replicates (2 pots per replicate), with 20 seedlings per replicate. The age of plants is indicated as days after the treatment imposition (DAT). The experiment lasted for 10 DAT, totaling 24 days after planting. Plant samples were collected at the initial time before treatment imposition (0 DAT), at 5 DAT, and 10 DAT. These plants were used to determine Fe (III) reductase activity (FCR, five plants per replicate) and for gene expression analysis (1 plant per replicate), as described below. SPAD index was measured at 0, 2, 4, 6, 8, and 10 DAT. Plants used for biomass, root morphological parameters, and elemental analysis were collected only at the end of the experiment (10 DAT), using 2 plants per replicate.

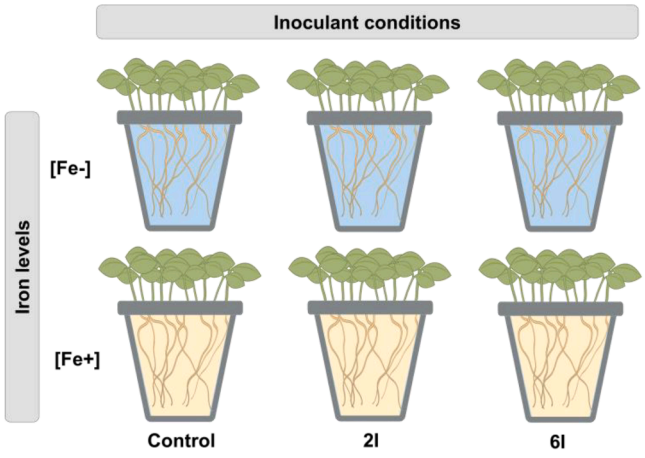


Fig. 1. The schematic diagram of the experimental design of trial 1. Plants were hydroponically grown on a nutrient solution without (Fe-) or with (Fe+) supplemented as Fe-EDTA and three inoculant conditions with *Pseudomonas* RMC4 (root inoculation 2 DAP (2I), root inoculation 6 DAP (6I), and non-inoculated control plants). Created with BioRender (https://biorender.com/).

2.5. Determination of Fe (III) reductase activity

Quantification of the reduction of Fe(III)-EDTA (FCR activity), was conducted on the root system of hydroponically cultured cucumber plants using the bathophenantroline disulfonate (BPDS) colorimetric method, as previously described (Vizzotto et al., 1999). In brief, roots of intact plants were submerged in a reagent solution composed of 0.5 mM $CaSO_4$, 10 mM MES NaOH (pH 5.5), Fe(III)-EDTA 0.25 mM, and BPDS 0.6 mM incubated in the dark at 25 °C. Following an incubation period of 30 min, the absorbance was measured at 535 nm. Subsequently, the amount of Fe(III) reduced was quantified based on the formation of Fe (II)-$BPDS_3$ complex, employing a molar extinction coefficient of 22.1 mM^{-1} cm^{-1}.

2.6. Plant analysis

2.6.1. SPAD index

The leaf chlorophyll index was assessed in plants from trial 2 at two-day intervals after the treatment imposition (0, 2, 4, 6, 8, and 10 DAT) employing a digital chlorophyll meter SPAD 502 (Minolta Co., Osaka, Japan). Measurements were conducted within each of the three biological replicates, on three individual plants per pot. A minimum of two readings were recorded per pot and averaged.

2.6.2. Plant biomass

Two plants per replicate of each treatment were taken at the harvest. Plants were separated into roots and shoots and oven-dried at 65 °C to constant weight. Subsequently, the dry weight (DW) was determined for each of the plant tissues.

2.6.3. Root morphological characteristics

For the analysis of root morphology, two fresh roots per replicate of each treatment were scanned using a root scanner system (EPSON Perfection V800, Regent Instruments Inc., Quebec, Canada), and length, surface area, diameter, volume, tips, forks, and crossings, were determined and analyzed with the WinRHIZO software (EPSON 1680, WinRHIZO Pro2003b).

2.7. Ionomic analysis

Dry shoots and roots were digested with 68 % ultrapure HNO_3 (Carlo Erba, Milano, Italy) in a single reaction chamber (SRC) microwave digestion system (UltraWAVE, Milestone, Shleton, CT, USA). Concentrations of macro and micronutrients were obtained by Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES, Arcos Ametek Spectro, Germany).

2.8. Gene expression analysis

Root samples for gene expression analysis were collected and immediately frozen in liquid nitrogen and stored at −80 °C until use. Total RNA was extracted from cryopreserved root samples employing the Spectrum Plant Total RNA Kit (Sigma-Aldrich, St, MO, USA), according to the manufacturer's specifications. Subsequently, 1 μg of total RNA was treated with DNase (RQ1, 10 U) and used for cDNA synthesis using the ImProm-II Reverse Transcription System (Promega, Madison, WI, USA) and oligo $(dT)_{15}$ primer, as specified by the manufacturer. The

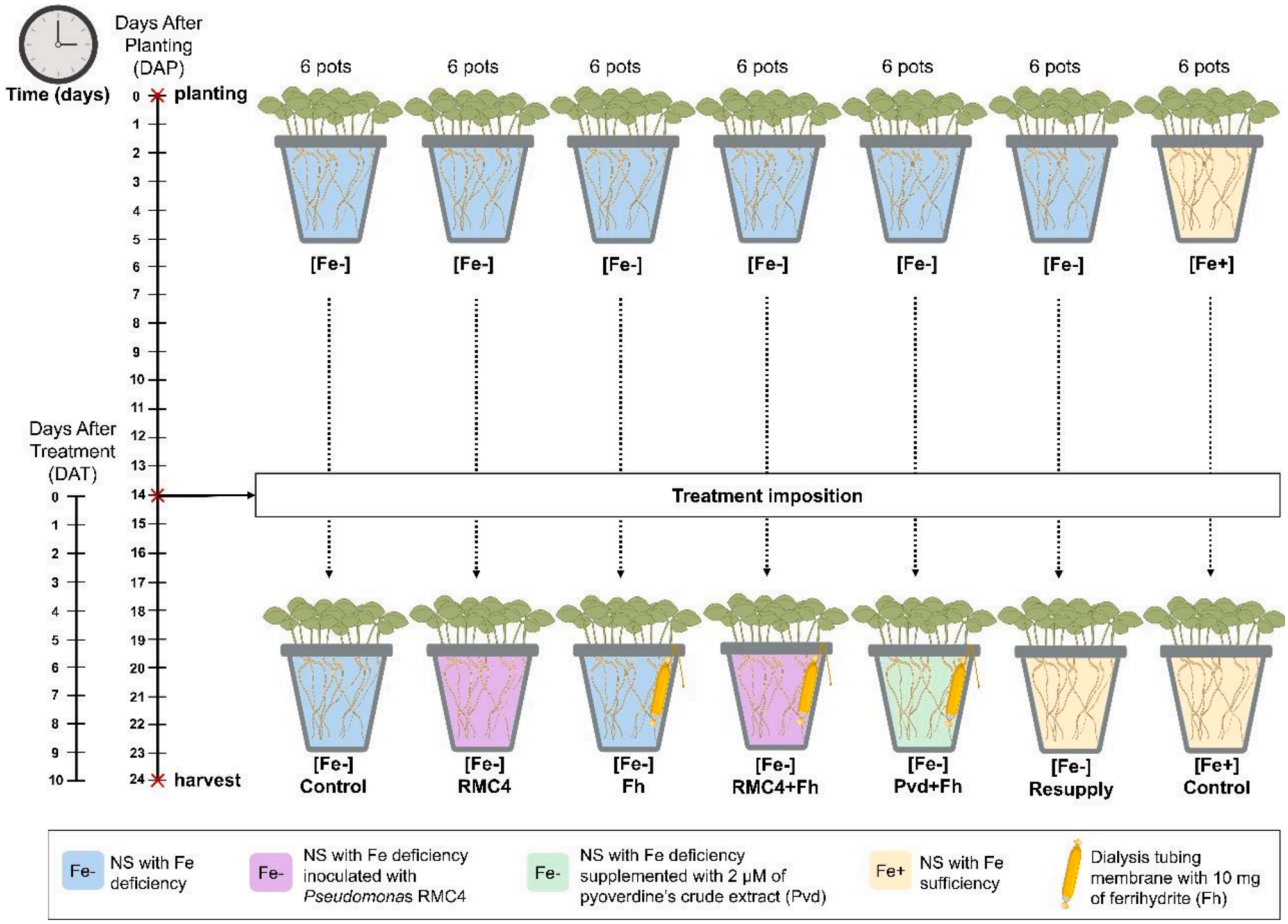


Fig. 2. Timeline of plant cultivation and experimental design of trial 2. Cucumber seedlings were grown on a nutrient solution (NS) without iron (Fe-) or with iron (Fe+, supplied as 80 µM Fe-EDTA). Fourteen days after planting (14 DAP), plants were divided according to the treatment imposed: [Fe-] Control, [Fe-] RMC4, [Fe-] Fh, [Fe-] RMC4+Fh, [Fe-] Pvd+Fh, [Fe-] Resupply, and [Fe+] Control. Plants were grown for 10 more days after treatment imposition (10 DAT). Created with BioRender (https://biorender.com/).

resulting cDNA served as the template for quantitative real-time reverse transcription PCR (qRT-PCR), which was carried out using the SsoFast EvaGreen Supermix (Bio-Rad, Hercules, CA, USA) and the Bio-Rad iCycler MyiQ real-time PCR system (Bio-Rad). Target genes and the housekeeping gene were amplified using gene-specific primers, as previously described by Pii et al. (2016) and Marastoni et al. (2019). For each transcript, the mean normalized expression value (MNE; Simon, 2003) was calculated using the housekeeping transcript, and the relative expression ratio values were calculated by the $2^{-\Delta\Delta Ct}$ method according to (Livak and Schmittgen, 2001).

2.9. Data analysis

Data were expressed as means of three replicates ± standard errors (SE). One-way analysis of the variance (ANOVA) was performed using R software (version 2023.12.1 + 402) and the mean values were separated according to Scott-knott test with $p < 0.05$. The following R packages were used for data visualization and statistical analyses: ggplot2, agricolae, ggpbur, and ScottKnott.

3. Results

3.1. Trial 1: effect of iron nutrition and Pseudomonas inoculation on cucumber growth and Fe uptake mechanisms

3.1.1. Plant biomass

After 10 days of hydroponic growth, cucumber plants were influenced by the different treatments (Table 1). Regardless of bacterial inoculation, the root dry weight (RDW) of plants grown under Fe-deficient conditions was significantly higher than Fe-sufficient plants. Moreover, when considering early bacterial inoculation (2I), its effect in enhancing root accumulation was more evident under Fe starvation than late inoculation (6I), compared to uninoculated control plants, whereas no significant effects were induced by the bacteria under optimal Fe conditions (Table 1). On the other hand, the shoot dry weight (SDW) was higher in Fe-sufficient plants, yet no significant effect was induced by inoculation treatments. However, early inoculation (2I) significantly enhanced SDW by 28 % in Fe-starved plants.

Table 1
Root dry weight (RDW) and shoot dry weight (SDW) of cucumber plants grown under Fe-deficient (Fe-) or Fe sufficient (Fe+) conditions and inoculated with *Pseudomonas* RMC4, at the end of the experiment (10 DAP).

Biomass (mg)	**Inoculant treatments with *Pseudomonas* RMC4**					
	Fe deficiency [Fe-]			Fe sufficiency [Fe+]		
	Control	2I	6I	Control	2I	6I
RDW	6.40 b	7.68 a	6.49 b	5.38 c	5.23 c	5.18 c
SDW	46.5 b	59.5 a	50.1 b	56.8 a	56.6 a	54.9 a

Differences between means were determined using Scott-Knott test ($p < 0.05$). The data are reported as means ($n = 3$). Different letter in the same row indicates significant differences according to Scott-Knott test.

3.1.2. Ferric-chelate reductase (FCR) activity

The time course of FCR activity in the roots of cucumber plants is shown in Fig. 3. Considering plants grown under Fe-deficient conditions (Fe-), the early inoculation treatment (2I) induced a significant improvement in the FCR ability on 6-day-old plants (i.e., at four days after inoculation) with respect to non-inoculated plants (Control). Similarly, when the inoculation was carried out on 6-day-old plants (6I) that already express FCR activity, a significant increase in the magnitude of the Fe-reduction capacity was observed three days after inoculation (i.e., on 9-day-old plants) compared to control plants (Fig. 3). When considering Fe sufficiency conditions (Fe+), the FCR activity showed a significant upregulation, starting from day 8 until the end of the experiment, in early-inoculated plants (Fig. 3 and Supplementary Table 1). In late inoculated plants, FCR has never displayed significantly different values in comparison to non-inoculated control plants. Noteworthy, Fe availability strongly influenced FCR activity in the roots of cucumber. Thus, while the FCR activity of plants grown under Fe sufficiency ranged from 50 to 150 nmol Fe^{2+} h^{-1} g^{-1} fresh weight, the Fe-deficient plants showed FCR values ranging from 1500 to 2000 nmol Fe^{2+} h^{-1} g^{-1} fresh weight.

3.1.3. Gene expression analysis

Based on the most significant effects induced by the inoculation with *Pseudomonas* RMC4 during the time course FCR activity, the 2-, 6-, and 10-old-day plants grown under Fe deficiency, as well as, the 2- and 8-old-day plants grown under Fe sufficiency were used to analyze the transcriptional regulations of genes encoding the Fe-chelate reductase (i.e., *CsFRO1* and *CsFRO3*, Fig. 4). The relative expression ratios were calculated using non-inoculated control 2-day-old plants as calibrator. The results showed that Fe deficiency induced a high overexpression of *CsFRO1* along the time (Fig. 4A). However, the early inoculation ([Fe-] 2I) did not induce significant expression of this gene when compared to non-inoculated control plants, whereas late-inoculated Fe starved plants (([Fe-] 6I) showed a significantly lower *CsFRO1* expression at 10 days (Fig. 4A). In addition, Fe starvation also induced the overexpression of *CsFRO3* along the 10 days of growth (Fig. 4B). Nonetheless, differently than *CsFRO1*, both early and late inoculation induced the higher expression of *CsFRO3* (1.2-fold and 1.8-fold, respectively) when plants were 10-day-old, compared to control plants at the same growing time. The distinctive regulation of *CsFRO1* and *CsFRO3* suggest a different gene expression depending on both Fe nutritional status and the presence of the bacterium. On the other hand, Fe sufficient control plants showed a down-regulation of *CsFRO1* after 8 days of growing (4.3-fold), when compared to plants at 2 days (Fig. 4C). Nonetheless, plants early inoculated (2I) showed a higher induction of *CsFRO1* (8-fold) at six days after inoculation (i.e., 8 days of growing) as compared to control plants at the same time, whereas late-inoculated plants (6I) slightly increased the expression of *CsFRO1* (1.6-fold).

3.2. Trial 2: effect of Pseudomonas and its siderophore on iron-starved cucumber plants

3.2.1. Plant growth and root morphological parameters

Plant biomass accumulation was clearly affected by the treatments imposed (Table 2). As expected, plants adequately fed with Fe during the whole experiment ([Fe+] Control) presented the highest root and shoot biomass accumulation as compared to the other treatments. Nonetheless, when considering plants growing under Fe starvation, those that were treated with *Pseudomonas* in combination with ferrihydrite ([Fe-] RMC4+Fh) showed a RDW 15 % higher than non-inoculated control

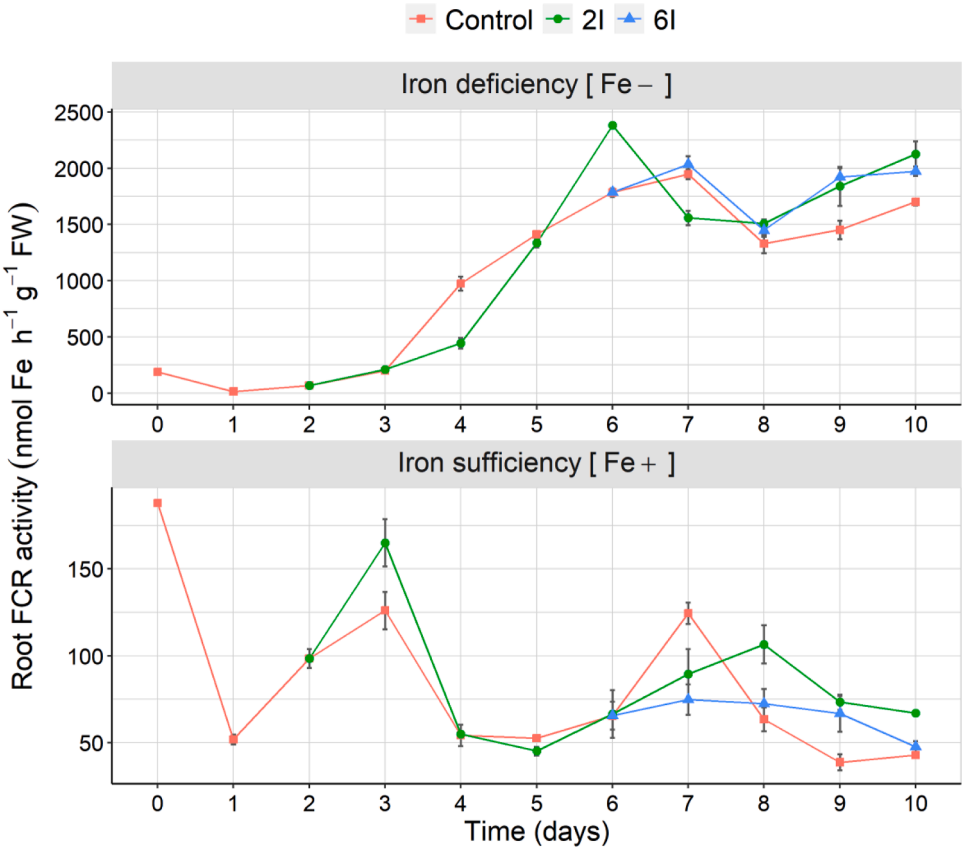


Fig. 3. Time course of ferric-chelate reductase (FCR) activity determined in roots of cucumber plants grown hydroponically in either Fe deficiency (Fe-) or Fe sufficiency (Fe+). Plants were early inoculated at day 2 (2I), late inoculated at day 6 (6I), or non-inoculated (Control) with *Pseudomonas* RMC4. Data reported are the means ± SE, $n = 3$.

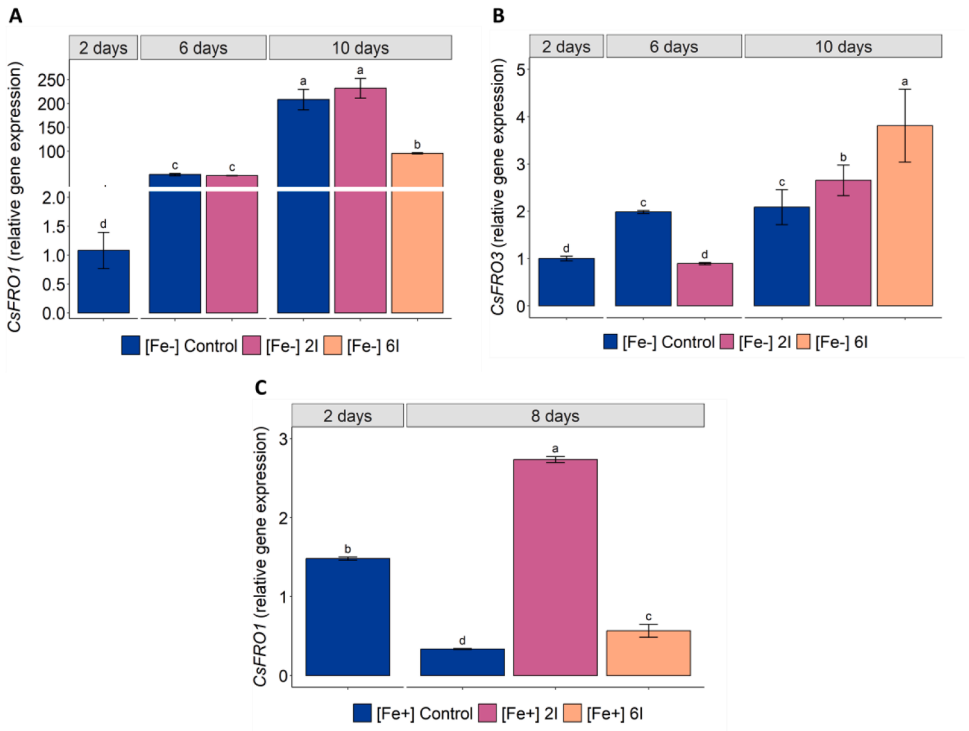


Fig. 4. Time course expression analysis of *CsFRO1* and *CsFRO3* in roots of cucumber plants grown either in Fe deficiency (A, B) or Fe sufficiency (C) conditions and either non-inoculated or inoculated with *Pseudomonas* RMC4 at 2 (early inoculation) and 6 (late inoculation) days. The expression level of each gene was normalized to the expression level of the *elongation factor isoform 1-alpha* (EF-1α). The relative expression ratios were calculated using non-inoculated control 2-day-old plants as a calibrator sample. The values reported are means ± SE ($n = 3$). Different letters on the same graph showed statistical differences according to Scott-Knott test ($p < 0.05$).

Table 2
Mean values for the plant biomass and root morphological parameters of Fe-deficient cucumber plants subjected to different treatments at the end of the experiment (24 DAP).

Treatments	Biomass		Root morphology			
	RDW (mg · plant^{-1})	SDW (mg · plant^{-1})	Length (cm)	Surface area (cm^2)	Diameter (mm)	Volume (cm^3)
[Fe-] Control	5.2 c	93.9 c	194.1 d	17.7 d	0.29 a	0.13 d
[Fe-] RMC4	7.3 b	121.0 c	293.8 c	25.3 c	0.31 a	0.20 c
[Fe-] Fh	5.5 c	96.9 c	211.0 d	19.3 d	0.29 a	0.14 d
[Fe-] RMC4+Fh	6.0 c	107.8 c	190.2 d	17.4 d	0.29 a	0.13 d
[Fe-] Pvd+Fh	3.5 d	79.7 c	167.4 d	10.8 e	0.22 b	0.06 e
[Fe-] Resupply	5.2 c	187.4 b	360.0 b	32.6 b	0.29 a	0.24 b
[Fe+] Control	12.9 a	302.8 a	592.5 a	53.9 a	0.29 a	0.39 a

Differences between means were determined using Scott-Knott test ($p < 0.05$). The data are reported as means ($n = 3$). Different letter in the same column indicates significant differences according to Scott-Knott test.

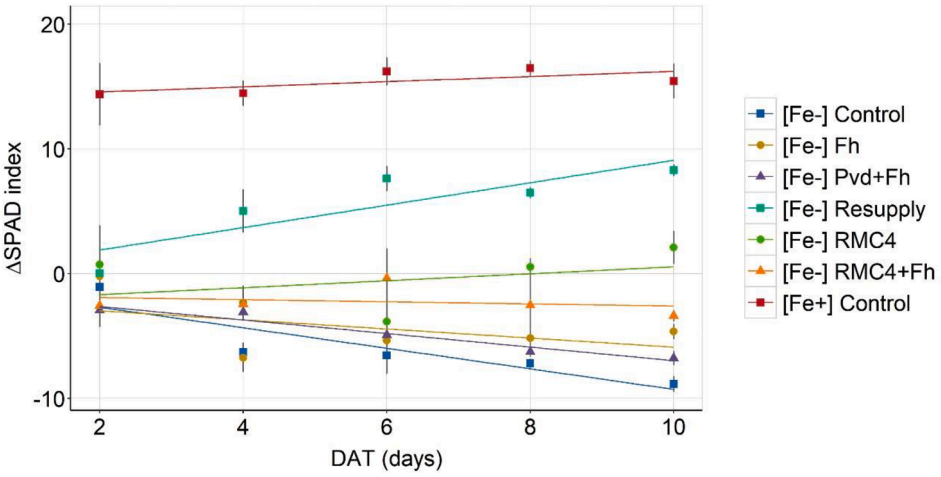
plants ([Fe-] Control). However, it was the inoculation with the bacterium alone ([Fe-] RMC4) that allowed the higher increase in RDW (by 40 %) compared to [Fe-] control plants (Table 2). Conversely, the addition of pyoverdine's crude extract in combination with Fh ([Fe-] Pvd+Fh) caused a significant reduction in root biomass (by 33 %) as compared to untreated control plants. Moreover, although the shoot biomass data did not show any significant differences among the treatments under Fe-starvation, those in which *Pseudomonas* was applied presented higher values than untreated control plants. On the other hand, plants supplied with Fe after a previous Fe-starved growth ([Fe-] Resupply) doubled their SDW compared to [Fe-] control plants (Table 2).

In terms of root morphology (Table 2), plants grown under optimal Fe conditions ([Fe+] Control) also exhibited higher values in all assessed parameters, followed by plants supplied with Fe after a period of Fe deficiency (14 DAP) ([Fe-] Resupply). However, among the treatments subjected to Fe starvation, inoculation with *Pseudomonas* RMC4 resulted in the highest values, increasing root length, surface area, and volume by 51 %, 43 %, and 54 %, respectively, compared to untreated control plants ([Fe-] Control). In contrast, the addition of pyoverdine in combination with Fh ([Fe-] Pvd+Fh) yielded the lowest values, decreasing root length, surface area, diameter, and volume by 14 %, 39 %, 24 %, and 54 %, respectively, when compared to Fe-deficient control plants (Table 2).

3.2.2. SPAD

The variation of SPAD index over time indicated the effect of treatments in the re-greening of cucumber leaves first exposed to a period of Fe deficiency and subsequently exposed to different treatments (Fig. 5). As expected, the most positive trend in the regression line (slope = 0.899 SPAD units per day) was observed in [Fe-] Resupply, i.e., plants first exposed to Fe deficiency and then resupplied with 80 μM of Fe-EDTA, thus showing a fast recovery from Fe deficiency symptoms (Fig. 5). On

A



B

Treatment	Line's equation	R^2	*p*-value	Scott-Knott
[Fe-] Control	$y = -0.823x - 1.039$	0.792	0.043	c
[Fe-] Fh	$y = -0.365x - 2.235$	0.216	0.430	c
[Fe-] Pvd + Fh	$y = -0.543x - 1.536$	0.944	0.006	c
[Fe-] Resupply	$y = 0.899x + 0.100$	0.745	0.059	a
[Fe-] RMC4	$y = 0.279x - 2.241$	0.131	0.549	b
[Fe-] RMC4 + Fh	$y = -0.084x - 1.744$	0.057	0.698	c
[Fe+] Control	$y = 0.206x + 14.158$	0.454	0.272	b

Fig. 5. Time course of chlorophyll content. (A) Values reported in the plot were expressed as variation of SPAD index (ΔSPAD), calculated as the difference between SPAD values at different days after treatments (DAT) and the values at the beginning (0 DAT). Treatments are listed as follows: Fe sufficiency ([Fe+] Control), Fe deficiency ([Fe-] Control) and treated either with Fe re-supply ([Fe-] Resupply), *Pseudomonas* RMC4 ([Fe-] RMC4), ferrihydrite ([Fe-] Fh), ferrihydrite and pyoverdine ([Fe-] Pvd + Fh), or ferrihydrite and *Pseudomonas* RMC4 ([Fe-] RMC4 + Fh). Data are reported as means ± SE, $n = 3$. (B) Table reporting the results of regression model shown in panel A, in particular line equation, R^2, and *p-values* referring to the regression model. The one-way ANOVA analysis with Scott-Knott post-test is referred to the values of the lines' slopes.

the other hand, the [Fe-] Control showed the most negative trend (slope = −0.823 SPAD units per day), showing therefore a worsening in the Fe chlorosis symptoms during the experimental period. Consistently, [Fe+] Control samples, i.e., always supplied with adequate amount of Fe in the nutrient solution, showed slightly positive trend in the regression line (slope = 0.206 SPAD units per day), suggesting slow increase in the concentration of chlorophyll over time. Interestingly, a similar trend (slope = 0.279 SPAD units per day) was also observed in [Fe-] RMC4 samples, possibly highlighting a positive effect of RMC4 inoculation on SPAD index, independently from the availability of Fe in the nutrient solution (Fig. 5). All the other treatments carried out, i.e., [Fe-] Fh, [Fe-] Pvd + Fh and [Fe-] RMC4 + Fh, showed a statistically similar outcome with respect to [Fe-] Control, thereby displaying no positive effects on the recovery from Fe deficiency symptoms, at least in the timeframe considered for the present experiment (Fig. 5).

3.2.3. *Iron chelate reductase (FCR) activity*

In order to further investigate the physiological responses of Fe deficient cucumber plants and subjected to the different treatments, the FCR activity was measured on DAT 0, 5 and 10 (Fig. 6). At 0 DAT, a significant difference was observed between the Fe sufficient plants ([Fe+] Control) and all the other treatments, in which cucumber plants have undergone a 14-day Fe starvation period. At 5 DAT, a general decrease in FCR activity was observed in all treatments, albeit [Fe-] Resupply showed the strongest decrease, whilst [Fe-] Control the mildest one (Fig. 6 and Supplementary Table 2). Interestingly, among the other treatments, those featuring either *Pseudomonas* RMC4 or pyoverdine in combination with ferrihydrite displayed the more pronounced decrease in the FCR (Fig. 6 and Supplementary Table 2). Similarly, also the treatments [Fe-] RMC4 and [Fe-] Fh displayed a reduction in FCR activity, albeit to a lower extent respect to the counterparts supplemented with ferrihydrite (Fig. 6 and Supplementary Table 2). When considering the FCR activity values detected at 10 DAT, the majority of the treatments showed stable values, i.e., no statistically significant differences were displayed with respect to those detected at 5 DAT (Fig. 6 and Supplementary Table 2). Nevertheless, [Fe-] RMC4 + Fh and [Fe-] Resupply samples showed a significant increase in the Fe reductase activity at 10 DAT (Fig. 6 and Supplementary Table 2).

3.2.4. *Gene expression analysis*

To further investigate the response of cucumber plants towards the different treatments imposed, root tissues were sampled at 0, 5 and 10 DAT and the expression of key genes was assessed. In particular, considering the data obtained previously (Marastoni et al., 2019; Pii

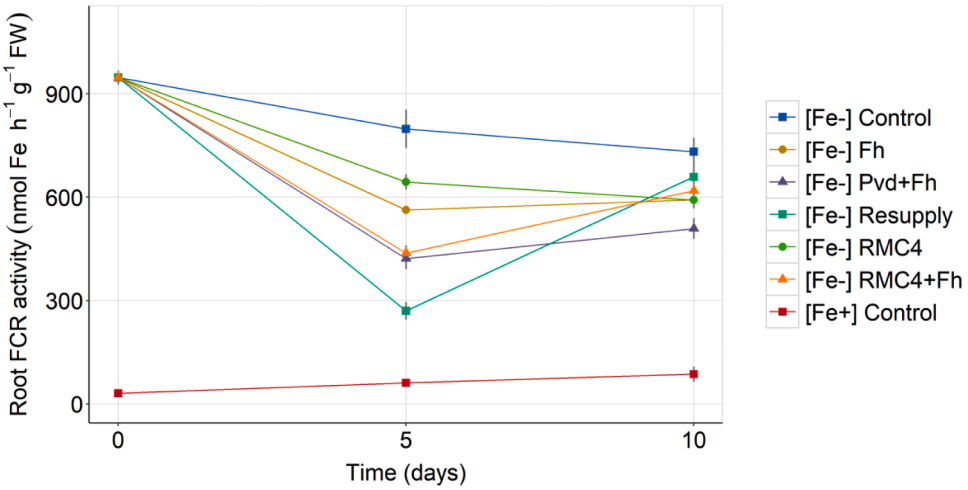


Fig. 6. Time course of Fe chelate reductase (FCR) activity**.** The FCR activity was determined in roots of hydroponically grown cucumber plants at 0, 5 and 10 days after treatments. Treatments are listed as follows: Fe sufficiency ([Fe+] Control), Fe deficiency ([Fe-] Control) and treated either with Fe re-supply ([Fe-] Resupply), *Pseudomonas* RMC4 **(**[Fe-] RMC4), ferrihydrite ([Fe-] Fh), ferrihydrite and pyoverdine ([Fe-] Pvd + Fh), or ferrihydrite and *Pseudomonas* RMC4 ([Fe-] RMC4 + Fh). Data are reported as means ± SE, $n = 3$.

et al., 2016), the transcriptional modulation of *CsFRO1* and *CsFRO3* was analyzed (Fig. 7**).** The relative expression ratios were calculated using untreated Fe deficient plants ([Fe-] Control) at 0 DAT as reference. Concerning *CsFRO1*, the data obtained confirmed that Fe starvation caused an up-regulation of the gene with respect to [Fe+] Control samples (Fig. 7). In addition, at 5 DAT, *CsFRO1* was further induced by Fe starvation in [Fe-] Control, whilst it was not significantly modulated at 10 DAT with respect to the initial values (Fig. 7). Similarly, in Fe deficient plants the treatment with *Pseudomonas* RMC4 and Ferrihydrite caused an upregulation of *CsFRO1* over time, reaching the highest expression levels at 10 DAT, as compared to [Fe-] Control at 0 DAT (Fig. 7). On the other hand, the supplementation of *Pseudomonas* RMC4 and pyoverdine in combination with ferrihydrite (i.e., [Fe-] RMC4 + Fh and [Fe-] PVD + Fh) caused a downregulation of *CsFRO1* at 5 DAT and its upregulation at 10 DAT (Fig. 7). Considering the expression of *CsFRO3*, it resulted not significantly modulated at 5 DAT in the treated plants. On the contrary, at 10 DAT, a significant upregulation of *CsFRO3* was observed in [Fe-] RMC4, [Fe-] Fh, [Fe-] RMC4 + Fh, [Fe-] PVD + Fh and [Fe-] Resupply. However, it is worth mentioning that *CsFRO3* expression levels, in all the treatments and the time points considered, was on average 90-fold lower as compared to *CsFRO1* expression levels (data not shown).

3.2.5. *Ionomic analysis*

The analysis of ionomic signature of cucumber plants has been carried out at the end of the experimental period (i.e., 10 DAT) by assessing the concentration of key microelements, such as Fe, manganese (Mn), zinc (Zn), and copper (Cu), and their relationships. Nutritional molar ratios (Fe/Mn and Fe/(Cu+Zn+Mn)) in shoot are shown in Table 3, whilst the concentration values of each micronutrient in different plant tissues are given in Supplementary Table 3. Concerning Fe/Mn ratio, as expected, the highest ratio was showed by [Fe+] Control plants, followed by resupplied Fe deficient plants (i.e., [Fe-] Resupply) and by [Fe-] Fh. Moreover, [Fe-] RMC4 and [Fe-] RMC4 + Fh showed similar Fe/Mn molar ratio. In addition, treatments with *Pseudomonas* RMC4 (i.e., [Fe-] RMC4 and [Fe-] RMC4 + Fh) showed significantly higher Fe/Mn molar ratios than the [Fe-] Pvd + Fh, suggesting a possible negative effect of the bacterial pyoverdine's crude extract on Fe uptake/translocation by the plant. In agreement with this observation, [Fe-] Pvd + Fh and [Fe-] Control plants did not show significant differences in Fe/Mn molar ratios.

As expected, a similar trend observed for Fe/(Mn+Cu+Zn) molar ratio (Table 3) suggesting that the Fe treatments applied in the present assay did not affect the plant uptake of the other micronutrients Zn and Cu.

4. Discussion

4.1. *Trial 1: Pseudomonas RMC4 enhanced Fe acquisition mechanisms in cucumber plants*

Along with other essential macro- and micro-nutrients, Fe nutrition represents one of the main constraints to achieve equilibrated crop development and optimal productivity (Kobayashi and Nishizawa, 2012). At present, in crop fertilizing programs Fe is mainly supplied to plants in the form of synthetic chelates, albeit serious concerns regarding these compounds have been raised (Zuluaga et al., 2021). In this context, the exploitation of PGPB, thanks to their contribution to the Fe soluble fraction in soil, might represent an interesting perspective. Based on evidence collected previously (Pii et al., 2016), a short-term experiment was designed to establish the effect of the bacteria *Pseudomonas* RMC4 on cucumber plants, grown in different Fe availability conditions (i.e., Fe deficiency [Fe-], and Fe sufficiency [Fe+]), and inoculated with the PGPB either 2 (2I) or 6 (6I) days after planting. The early inoculation procedure (2I) induced an increase in the accumulation of root and shoot biomass in Fe starved plants with respect to Fe sufficient ones, both non-inoculated and inoculated with *Pseudomonas* RMC4 (Table 1**).** It is worth noting that plants are known to continuously release in the rhizosphere organic compounds, such as amino acids, sugars, and organic acids, among others, generally indicated as root exudates (Vives-Peris et al., 2020). The qualitative and quantitative composition of exudates released by plants can change depending on different factors, for instance, the plant genotype, growing substrate (e.g., hydroponics vs. soil), environmental conditions, agricultural practices, and nutrients availability, to name a few (Cesco et al., 2021; Vives-Peris et al., 2020). In this respect, Carvalhais et al. (2011) demonstrated that Fe deficiency in maize plants can enhance the levels of glutamate, glucose, ribitol, and citrate in the released exudates. Consistently, cucumber plants grown in the absence of Fe were shown to enhance the release of both amino acids and organic acids, which, among other functions, are supposed to exert a chemoattractant activity (Pii et al., 2015). Moreover, under Fe deficiency conditions cucumber plants can secrete riboflavin and its derivatives (Bosch et al., 2024), the importance of these compounds in the establishment of an efficiency

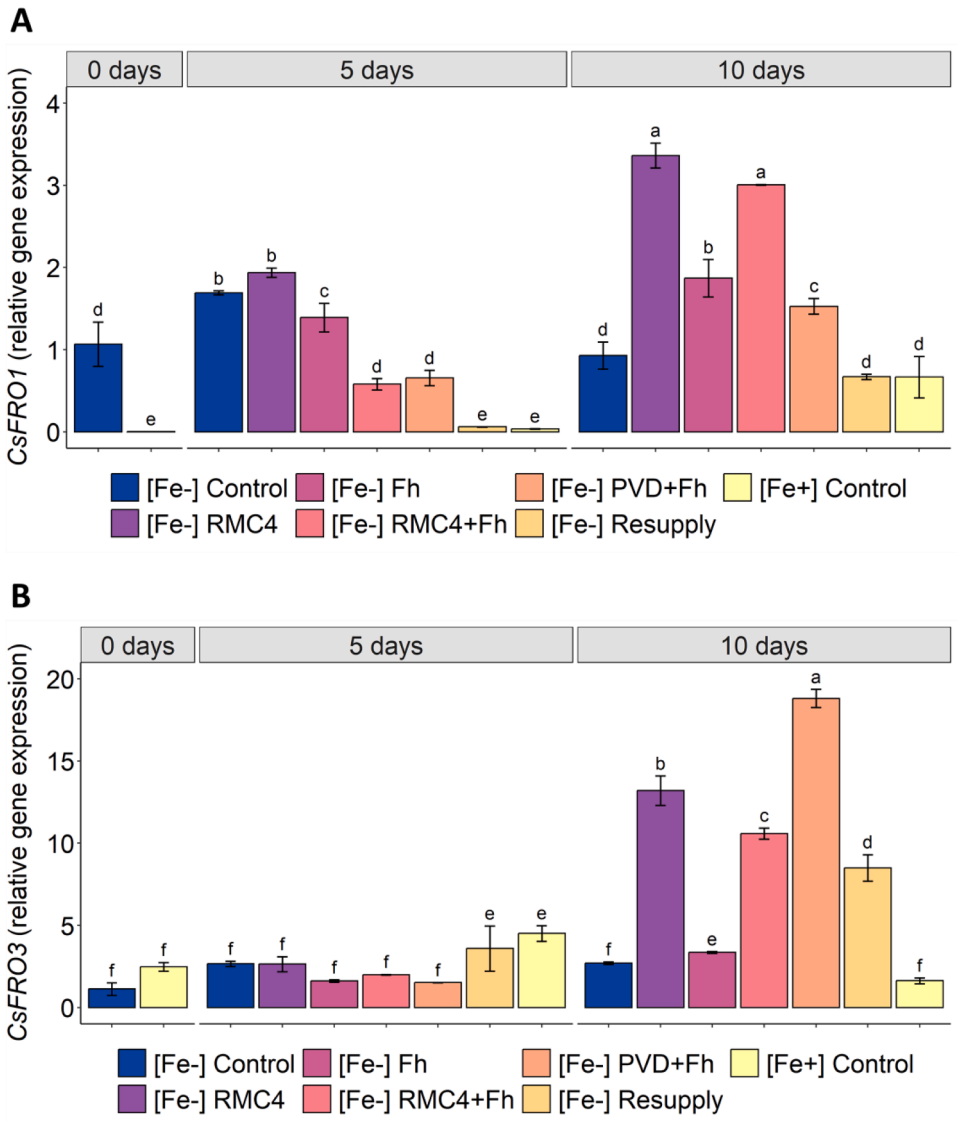


Fig. 7. Time course expression analysis of (A) *CsFRO1* and (B) *CsFRO3* in roots of cucumber plants subjected to different treatments. Treatments are listed as follows: Fe sufficiency ([Fe+] Control), Fe deficiency ([Fe-] Control) and treated either with Fe re-supply ([Fe-] Resupply), *Pseudomonas* RMC4 ([Fe-] RMC4), ferrihydrite ([Fe-] Fh), ferrihydrite and pyoverdine ([Fe-] Pvd + Fh), or ferrihydrite and *Pseudomonas* RMC4 ([Fe-] RMC4 + Fh). The expression level of each gene was normalized to the expression level of the *elongation factor isoform 1-alpha* (*EF-1α*). The relative expression ratios were calculated using [Fe-] Control plants at 0 DAT as a calibrator sample. The values reported are means ± SE ($n = 3$). Different letters on the same graph showed statistical differences according to Scott-Knott test ($p < 0.05$).

Table 3
Micronutrient molar ratios (Fe/Mn and Fe/(Mn+Cu+Zn)) in shoots of cucumber plants subjected to different treatments. Nutritional molar ratios were calculated from the micronutrients' concentrations expressed in mol g^{-1} dry weight.

Treatments	Micronutrients molar ratio	
	Fe/Mn	Fe/(Mn+Zn+Cu)
[Fe-] Control	1.16 d	0.32 c
[Fe-] RMC4	1.31 c	0.35 c
[Fe-] Fh	2.19 b	0.50 b
[Fe-] RMC4+Fh	1.55 c	0.40 c
[Fe-] Pvd+Fh	0.99 d	0.25 d
[Fe-] Resupply	2.21 b	0.55 b
[Fe+] Control	3.53 a	1.04 a

Differences between means were determined using Scott-Knott test ($p < 0.05$). The data are reported as means ($n = 3$). Different letter in the same column indicates significant differences according to Scott-Knott test.

symbiotic relationship between plant and bacteria has been reported (G. Yang et al., 2002). All the above-mentioned changes might have induced a positive effect on the interaction between *Pseudomonas* RMC4 and the host plant, enhancing its plat growth-promoting effect. In addition, in the experiment conducted by Hernández-Montiel et al. (2017), the efficiency of two inoculation methods (microcapsules and liquid inoculation) of different strains of *Pseudomonas putida* on the growth and yield of tomato plants was tested. Similar to the present experiment, an increase in dry weight was observed in treatments with bacterial inoculation via microcapsules or a bacterial mixture via liquid inoculation compared to the control treatment. In this context, it was suggested that the increase in dry weight was probably due to the production of IAA by the *Pseudomonas putida*. In this respect, Lozano-González et al. (2023) showed that the *Pseudomonas* RMC4 also produces IAA, which might account for the observed enhancement in biomass accumulation. Therefore, it is very likely that changes in the root exudation pattern induced by Fe shortage, in combination with the *Pseudomonas* RMC4-dependent IAA production, synergistically led to the observed

phenotypic alterations. However, inoculation at a later stage (6I) did not result in a significant increase in biomass, suggesting that the PGP effect requires time to manifest in the host plants.

Under conditions of Fe deficiency, cucumber plants activate their Fe acquisition mechanism. One aspect of this response is the increased expression levels of ferric chelate reductases (Jeong and Connolly, 2009), particularly the *FRO1* gene (Waters et al., 2007). Previous studies demonstrated that the PGPB *Azospirillum brasilense* had a positive effect on Fe deficient cucumber plants anticipating the activation of Fe acquisition mechanism (i.e., increase of the FCR activity and over-expression of *CsFRO1* gene), irrespective of their Fe nutritional status (Pii et al., 2016). The results regarding FCR indicated that *Pseudomonas* RMC4 inoculation indeed had an effect compared to non-inoculated Fe deficient plants on the last day of the experiment (10 DAT) (Fig. 3). However, no significant changes were observed in the expression of the *CsFRO1* and *CsFRO3* genes (Fig. 4A). Indeed, in the last few decades, the reductive properties of root exudates released by Fe deficient plants have been postulated (see review by Cesco et al., 2010; Schmidt, 1999). More recently, Fe-reducing coumarins and flavins have been demonstrated to be released by Fe deficient *A. thaliana* and *Beta vulgaris*, respectively (Schmid et al., 2014; Schmidt et al., 2014; Sisó-Terraza, et al., 2016a; Sisó-Terraza et al., 2016b). Consistently with Pii et al. (2016), also under optimal Fe conditions, the early inoculation of cucumber with *Pseudomonas* RMC4 caused an increase in the FCR activity sustained by an enhanced expression of *CsFRO1*. Interestingly, in this case, the biochemical and molecular modulations of plant responses following *Pseudomonas* RMC4 are postponed with respect to those observed with *A. brasilense* (Pii et al., 2016), possibly suggesting differences in the molecular plant-PGPB interactions. Indeed, the interaction between microbes and their hosts is a complex and multifaceted relationship that can yield varied results depending on the plant genotype and PGPB strain (Tripathi et al., 2024).

Overall, the results of Trial 1 demonstrated a biostimulant effect of the *Pseudomonas* RMC4 on cucumber plants, regardless of the Fe availability in the growth medium. Nevertheless, common (i.e., upregulation of Fe acquisition mechanisms) and peculiar (i.e., increase in the dry biomass in Fe deficient plants) responses have been observed.

4.2. Trial 2: exogenous application of Pseudomonas RMC4-derived pyoverdine improved fe acquisition but led to inadvertent stunted growth

Understanding the interactions between PGPB, its derived siderophores and sparingly soluble Fe sources can elucidate the mechanisms underlying Fe acquisition and utilization by plants. Several strains of *Pseudomonas* have been described for their ability to enhance plant growth and produce a wide diversity of siderophores, which can affect the levels of Fe availability in the rhizosphere for plants (Gao et al., 2022; Trapet et al., 2016). Under the experimental conditions adopted, the inoculation with *Pseudomonas* RMC4 was the most effective treatment in enhancing plant biomass under Fe starvation. However, the exogenous addition of pyoverdine's crude extract induced the opposite effect, limiting the plant biomass accumulation over the experimental period. Indeed, in the rhizosphere, PGPB-produced metabolites, like siderophores, may function synergistically with other microbial and plant metabolites, exerting positive effects on nutrient availability, stress tolerance, and growth promotion (Lucke et al., 2020). Nevertheless, in this work, the application of pyoverdine consisted in supplying a crude extract rather than a purified pyoverdine. Consequently, the presence of other bacterial metabolites in the extract cannot be excluded. In this respect, reports have shown that when certain metabolites are exogenously applied to plants, the delicate balance of biochemical pathways and biogeochemical cycles of nutrients could be altered, possibly leading to unintended toxicity symptoms, displayed for instance as stunted growth (Zuluaga et al., 2023b). Moreover, in the present conditions, the higher effectiveness of *Pseudomonas* RMC4 in boosting root development might be attributed to its ability of IAA synthesis. In fact, the auxins produced by beneficial *Pseudomonas* strains are known to stimulate changes in the root system architecture (Zamioudis et al., 2013).

It is well known that Fe deficiency leads to interveinal chlorosis, particularly in young leaves. Therefore, the decrease in SPAD values is a typical indication of Fe starvation (Hirai et al., 2007). Notably, within our experimental setup, both the inoculation of *Pseudomonas* RMC4 and the resupply with Fe-EDTA ([Fe-] Resupply) led to a noticeable increase of SPAD index values throughout the 10-day treatment period. Moreover, although not statistically significant, the addition of ferrihydrite and pyoverdine showed a higher recovery rate in comparison to the control group. These results indicate a recovery from nutrient deficiency symptoms, compared to untreated Fe-deficient plants. Usually, when facing Fe deficiency, Strategy I plants can release protons (H^+) and reducing substances (e.g., phenolics) through the root system (Santi and Schmidt, 2009). This phenomenon enhances the dissolution of sparingly soluble iron compounds, such as ferrihydrite, increasing the availability of iron in the rhizosphere. As a result, the plant's nutritional status improves, benefiting its overall metabolism and photosynthetic efficiency (Kobayashi et al., 2019). Additionally, PGPB strains, such as *Pseudomonas* spp., possess the ability to solubilize and mobilize Fe in the rhizosphere through the release of siderophores and organic acids, resulting in increased biosynthesis of photosynthetic pigments in leaves of Fe-deficient plants (Trapet et al., 2016).

The activity of FCR enzyme has been extensively used to comprehend the effectiveness of Strategy I in dicots plants in acquiring Fe under conditions of limited Fe-availability (Arcas et al., 2024). In this work, cucumber plants effectively responded to Fe deficiency by activating FCR over a 14-day period of starvation. However, decreased FCR rates were observed following either the bacterial inoculation, the application of pyoverdine or that of ferrihydrite, suggesting that such treatments increased the cucumber's ability to recover from Fe chlorosis. In fact, some studies have pointed out that, in Fe-starved plants resupplied with Fe in different forms (e.g., forms of ferric Fe, PGPB inoculation, Fe-chelating compounds), the FCR activity gradually decreases over time, approaching rates similar to those detected in Fe-sufficient conditions (Nagata et al., 2013; Puglisi et al., 2021; Zuluaga et al., 2023a). When ferrihydrite was added, the uptake of Fe by cucumber plants may be connected to the plant's inherent strategy, which involves proton secretion, acidification of the rhizosphere, and subsequent reduction of Fe^{3+} to more soluble ferrous Fe^{2+} by the FCR enzyme (Kobayashi et al., 2019). Regarding the combination of pyoverdine with ferrihydrite, it has been suggested that siderophore-mediated Fe uptake in Strategy I plants occurs through an as-yet-unidentified pathway not associated with Strategy I (Shirley et al., 2011; Vansuyt et al., 2007). In this regard, different authors have shown that different Fe acquisition strategies (e.g., Strategy I and II) might not be mutually exclusive, as demonstrated for tomato, Arachis hypogea and rice (Astolfi et al., 2020; Kobayashi and Nishizawa, 2012; Xiong et al., 2013).Moreover, in the case of inoculation with *Pseudomonas* RMC4, besides its proven ability to produce the pyoverdine siderophore, PGPB strains among this genus can also efficiently promote rhizosphere acidification by releasing organic acids, thereby reducing the pH of the medium and activating FCR in the roots (Roriz et al., 2021).

Analysis of the relative mRNA levels of *CsFRO1* transcripts at 5 DAT revealed a notable decrease in FCR rates of Fe-starved plants treated with *Pseudomonas* RMC4 or pyoverdine in combination with ferrihydrite. These findings suggest that the suppression of *CsFRO1* expression is a consequence of Fe restoration induced by *Pseudomonas* and its siderophore in the presence of sparingly soluble sources of Fe. Indeed, several authors have pointed out that, in Fe starved dicots, a decrease in the expression levels of *FRO* genes following a resupply with Fe sources serves as an indicator of plant recovery from Fe deficiency (Fuentes et al., 2018; Nagata et al., 2013). It is worth noting that PGPB themselves do not directly supply Fe to plants, but they have been shown to modulate the expression of *FRO* genes by altering the availability of Fe in

the rhizosphere *via* solubilization of insoluble Fe compounds and chelation of Fe ions (Rahimi et al., 2020). Hence, the increased availability of Fe triggers feedback mechanisms in plants that regulate *FRO* gene expression. Additionally, *FRO* genes in Fe-deficient plants were observed to partner with *IRT* (Fe-regulated transporter protein) and *FRU* (a putative transcription factor) genes being part of a local network cluster (CL:28,152) involved in siderophore-dependent Fe import into the cell. The biological processes associated with this gene cluster include Fe ion transmembrane transport, Fe ion transport, Fe ion homeostasis, and response to a bacterium (Kabir et al., 2021).

Iron deficiency in plants can disrupt the balance of other nutrients due to their interdependent interactions (Briat et al., 2015; Tomasi et al., 2014). It is well known that Fe homeostasis is often challenged by competition with other transition metals, including Cu, Zn, and Mn during uptake, transport, and chemical reactions within plant cells (Rai et al., 2021). For all the above mentioned, the content of mineral micronutrients of plants has been evaluated through the nutritional molar ratios, i.e., Fe/Mn and Fe/(Mn+Zn+Cu), in shoots. Several authors have used the Fe/Mn molar ratio to assess the effectiveness of Fe chelates in plants recovering from Fe chlorosis. This ratio is useful because increased Fe uptake can reduce Mn uptake by plants (Nadal et al., 2009). Interestingly, the treatment with ferrihydrite ([Fe-] Fh) resulted in a high Fe/Mn molar ratio, indicating that the plant effectively solubilized and acquired Fe from the ferrihydrite. However, this effect was not observed in the [Fe-] RMC4 + Fh treatment. Probably, despite the biostimulant effect that the *Pseudomonas* RMC4 had on the plant (increased biomass, increased root growth among others), it also competed with the plant for Fe acquisition. This competition likely decreased the amount of Fe available to the plant, thereby lowering the Fe/Mn molar ratio compared to the corresponding treatment without bacteria ([Fe-] Fh). On the other hand, the treatment with *Pseudomonas* alone (i.e., [Fe-] RMC4) resulted in a higher Fe/Mn molar ratio respect to [Fe-] Control plants. Although both treatments lacked exogenous Fe, the higher Fe/Mn ratio observed in *Pseudomonas* treatment might be ascribable to *Pseudomonas* RMC4 enhancing Fe translocation within the plant. Surprisingly, [Fe-] Pvd + Fh showed the lowest Fe/Mn molar ratio. Pyoverdine is known for its ability to solubilize Fe from sparingly soluble sources (Robin et al., 2008). Therefore, it was expected that pyoverdine could solubilize Fe from ferrihydrite by forming a Pvd-Fe complex, and, thereby, making Fe available for plant uptake. Alternatively, even though pyoverdine had no effect, the plants' Fe acquisition mechanisms should have increased Fe concentration in plants' tissues, thus leading to an increased Fe/Mn molar ratio. However, neither of these two scenarios has been observed. As previously mentioned, some reports indicate that the exogenous addition of bacterial metabolites, as in the case of bacterial pyoverdine crude extract, could disturb the delicate balance of biochemical pathways and biogeochemical cycles of mineral nutrients (Zuluaga et al., 2023b). On the other hand, while pyoverdine may have solubilized Fe and formed a Pvd-Fe chelate, the plant might not have been able to take up this Fe. This could be due to a biochemical imbalance in the plant, as mentioned above, or to the inability of the Pvd-Fe chelate to effectively release Fe to the plant. Further studies would be needed to discern if Pvd-Fe chelate in a purer chemical form could be more effective in providing Fe to the plant. Finally, the nutritional molar ratio Fe/(Mn+Zn+Cu) was also evaluated to observe the influence of the applied treatments on Cu and Zn micronutrients. The trend observed was comparable as that of the Fe/Mn molar ratio, therefore the applied treatments had no significant undesirable influence on acquisition and translocation of the mentioned micronutrients.

5. Conclusions

The imperative to maximize crop yield while minimizing environmental degradation motivates the exploration of alternatives to synthetic Fe chelates. This research provides evidence that inoculating crops with PGPB mitigates the effects of Fe deficiency symptoms and promotes plant development. Although these bacteria do not completely solve the Fe deficiency issue, they stimulate the activation of Strategy I plants' inherent mechanisms for Fe uptake. For instance, in Trial 1, early inoculation of *Pseudomonas* RMC4 led to increased biomass accumulation and enhanced Fe acquisition mechanisms in cucumber plants. On the other hand, Trial 2 underscores the synergistic role of *Pseudomonas* RMC4 and its siderophore in efficiently activating Fe acquisition machinery, thus improving Fe availability, nutrient balance and, finally, the resilience of the plant against Fe shortage. Overall, these findings underscore the multifaceted nature of plant-microbe interactions and their impact on Fe acquisition and growth responses in cucumber plants. While *Pseudomonas* inoculation shows potential for enhancing root development and Fe uptake mechanisms, the magnitude of these effects can be influenced by various factors, including Fe availability and siderophore supplementation. Further research would be needed to elucidate the underlying molecular mechanisms and to optimize the application of microbial inoculants for improving plant Fe nutrition and productivity in favour of more resilient and sustainable agricultural systems.

Funding

This work was financially supported by the Italian Ministry of Education, University and Research (MIUR) through the project Cleopatra coded 2022AAATEA and by the Free University of Bolzano through the project COMPETITIVE. The authors also gratefully acknowledge the financial support of the Ministerio de Ciencia, Innovación y Universidades and the Agencia Estatal de Investigación through the project PID2022-141721OB-C2. José María Lozano-González is the recipient of the FPI grant from the Ministerio de Ciencia e Innovación (PRE-2019-091246).

CRediT authorship contribution statement

José María Lozano-González: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation. **Mónica Yorlady Alzate Zuluaga:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation. **Juan José Lucena:** Writing – review & editing, Supervision. **Sandra López-Rayo:** Writing – review & editing, Supervision. **Sonia Monterisi:** Writing – review & editing, Methodology, Investigation, Data curation. **Stefano Cesco:** Writing – review & editing, Supervision, Funding acquisition. **Youry Pii:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors declare that the data supporting the findings of this study are available within the paper and its Supplementary Information files. Should any raw data files be needed in another format they are openly accessible at e-cienciaDatos (https://doi.org/10.21950/VHPYHW).

Acknowledgments

The authors are grateful to the research group in Plant Physiology – Department of Biology, Universidad Autónoma de Madrid, for providing

the bacterial strain.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.stress.2024.100565.

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